REAL 1B – GOLD INFINITY / ROUTE A

A certified stronger lower sieve

on the original rough shifted-prime source

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Principal conclusion:

$$\liminf_{X\to\infty}\frac{A(X)}{\mathcal{N}_\psi(X)}\geq\frac{7}{2}\log 3-4\log 2-2\log\frac{12}{11}+\frac{61}{500}=1.020531534119\ldots.$$

Stronger source bound proved. ACCEPTANCE UNMET for G3+/G5. The lower coefficient is rigorously improved by 0.122. The requested level near 1.1, the zero-composite-discrepancy threshold 1.175141611174…, and twin-prime infinitude remain unproved. This is an application and extension of existing sieve machinery to the exact source; historical novelty is not established. It is not presented as a new parity-breaking theorem.

Abstract

We prove a stronger lower bound for the squarefree $X^{8/31}$-rough shifted-prime source without assuming a new prime-distribution conjecture. The mechanism combines a finite, adaptive Buchstab decomposition with the complete three-largest-factor distribution function in Pascadi's work. The coefficient adapter uses the boxed Iwaniec construction, including its small-prime component, rather than assuming that a restricted raw Rosser weight is factorable. A further switched upper bound retains the entire rough cofactor inside a bounded Type II coefficient and preserves the fixed residue -2. Its kernel is $(10/3)\omega((1-a-b-c)/c)$. An outward-rounded finite certificate proves an additional normalized mass exceeding 0.1220144061; we spend only the rational amount 61/500. All complementary terms are retained. The resulting conditional gap in the matched consumer decreases from 0.276610077055… to 0.154610077055…, without changing the still-open complete composite discrepancy.

1 The source, result, and exact acceptance scope

Fix $0\leq\psi\in C_c^\infty((1,2))$, with $I_\psi=\int\psi>0$. Set

$$\rho=\frac{8}{31},\quad z=X^\rho,\quad \mathfrak{S}_2=2\prod_{\ell>2}\left(1-\frac{1}{(\ell-1)^2}\right),\quad \mathcal{N}_\psi(X)=\mathfrak{S}_2 I_\psi\frac{X}{\log X}.$$

The physical source in this paper is exactly

$$A(X)=\sum_n\psi(n/X)\Lambda(n+2)\mu^2(n)\mathbf{1}_{\mathrm{P}^-(n)>X^\rho}. \tag{1.1}$$

No character replacement, approximate primality gate, or change of roughness is used.

Theorem 1.1 (Certified positive-source improvement)**.** *For the source* (1.1)*,*

$$\liminf_{X\to\infty}\frac{A(X)}{\mathcal{N}_\psi(X)}\geq a_*:=a_0+\frac{61}{500},\qquad a_0=\frac{7}{2}\log 3-4\log 2-2\log\frac{12}{11}. \tag{1.2}$$

The external analytic inputs are the unconditional BFI well-factorable distribution theorem, Iwaniec's factorable-remainder linear sieve, and Pascadi's Propositions 4.4, 5.2 and 5.5. Their precise uses are stated below. No complete lower-Rosser level above 4/7 *is assumed.*

The supplied matched consumer, retained at its inherited research-proof scope, is

$$P_1 = A - (B_2 + B_3) - \mathcal{E}_{\mathrm{comp}}, \qquad \frac{B_2 + B_3}{\mathcal{N}_\psi(X)} = c_* + o(1), \quad c_* = 1.175141611174\ldots. \tag{1.3}$$

Thus (1.2) reduces the remaining requirement to a matched bound

$$\limsup_j \frac{\mathcal{E}_{\mathrm{comp}}(X_j)}{\mathcal{N}_\psi(X_j)} < a_* - c_* = -0.154610077055\ldots$$

along unbounded scales, with a fixed margin. No such discrepancy estimate is proved here. In particular, even complete composite equidistribution, $\mathcal{E}_{\mathrm{comp}} = o(\mathcal{N}_\psi(X))$, would leave a negative reserve with this lower coefficient. The comparison constant is not recertified in this paper; Theorem 1.1 does not depend on it.

2 Sieve notation and external inputs

We first work with genuine prime outputs, using

$$\theta(m) = \log m\, \mathbf{1}_{m\in\mathbb{P}}, \qquad S(d;u) = \sum_{\substack{n:d|n\\(n/d,P(u))=1}} \theta(n+2)\psi(n/X), \qquad P(u) = \prod_{2<\ell<u} \ell.$$

All relevant inputs are odd. The finite exceptions at the prime 2 are negligible. Prefix primes are always greater than or equal to the cofactor sieve cutoff. In the direct prime-progression estimates the output satisfies $n + 2 \equiv 2 \pmod d$; in the switched output sieve the input satisfies $n \equiv -2 \pmod d$. These are two uses of the same physical shift $+2$.

Write f, F for the standard dimension-one linear-sieve functions, and use the normalized versions

$$\overline{f} = e^{-\gamma} f, \qquad \overline{F} = e^{-\gamma} F.$$

Their initial values and delay equations are

$$\overline{f}(s) = 0 \quad (s \le 2), \qquad \overline{F}(s) = 2/s \quad (0 < s \le 3), \tag{2.1}$$

$$\overline{f}(s) = 2\log(s-1)/s \quad (2 \le s \le 4), \tag{2.2}$$

$$(s\overline{f}(s))' = \overline{F}(s-1), \qquad (s\overline{F}(s))' = \overline{f}(s-1). \tag{2.3}$$

For an upper sieve with a requested cutoff above its relative level, sieve instead at the relative level. This gives the same expression $2/s$ for $0 < s < 1$ in (2.1). Both normalized functions tend to $e^{-\gamma}$.

The shifted-prime divisor density is $g(d) = 1/\varphi(d)$ for odd squarefree d, and

$$V(u) = \prod_{2<\ell<u} \left(1 - \frac{1}{\ell - 1}\right) \sim \frac{\mathfrak{S}_2 e^{-\gamma}}{\log u}. \tag{2.4}$$

Consequently, after summing a prime prefix of logarithmic exponents $t_1, \ldots, t_k$, a relative lower or upper sieve at total level X^D, cofactor cutoff X^w, has kernel

$$\frac{1}{w}\overline{f}\left(\frac{D - t_1 - \cdots - t_k}{w}\right) \quad \text{or} \quad \frac{1}{w}\overline{F}\left(\frac{D - t_1 - \cdots - t_k}{w}\right), \tag{2.5}$$

against the prime-prefix measure $\prod_i \mathrm{d}t_i/t_i$. These are *averaged prime-box estimates*, not pointwise asymptotics for each fixed large prefix.

2.1 The exact distribution classes

The BFI theorem gives arbitrary logarithmic saving for well-factorable weights at level $X^{4/7-\varepsilon}$ with fixed nonzero residue. The Iwaniec construction supplies lower and upper linear-sieve bounds with the usual f, F and a factorable remainder; see Lichtman [2, Theorem 2.10 and Section 5] for an explicit modern presentation.

Pascadi [3, Proposition 4.4] proves a convolution progression estimate for $MN \asymp X$, bounded coefficients α_n, β_m, where α is Siegel–Walfisz and supported on

$$P^-(n) \geq X^{1/(\log\log X)^3},$$

and three separated modulus factors satisfy, with fixed positive margins,

$$Q_1 \leq N/X^{\varepsilon}, \quad N^2 Q_2 Q_3^2 \leq X^{1-15\varepsilon}, \quad N^2 Q_2^5 Q_3^2 \leq X^{2-40\varepsilon}. \tag{2.6}$$

The progression error is the actual expression

$$\mathbf{1}_{mn\equiv h \pmod{q_1q_2q_3}} - \frac{\mathbf{1}_{(mn,q_1q_2q_3)=1}}{\varphi(q_1q_2q_3)}, \qquad h \neq 0 \text{ fixed}.$$

Proposition 5.2 factorizes the complete *upper* linear-sieve support at level $X^{3/5-\varepsilon}$ for $X^{2\delta} \leq N \leq X^{1/3+\delta}$. It does not give the same assertion for complete lower weights. The 5/8 theorem concerns triply-well-factorable weights.

2.2 The three-factor anatomy function

For $t \geq 0$ put

$$\vartheta(t) = \min\left(\frac{1+t}{2}, 1 - \frac{3}{2}t\right).$$

For $0 \leq c \leq b \leq a \leq 1/4$, define the limiting function $\Theta(a,b,c)$ as the maximum of

$$\vartheta(a),\ \vartheta(b),\ \vartheta(a+b),\ \vartheta(a+b+c), \tag{2.7}$$

$$w(a,b,c), \quad w(b,a,c), \tag{2.8}$$

$$\vartheta(a+c)\mathbf{1}_{\vartheta(a+c)>a+2b+c}, \quad \vartheta(b+c)\mathbf{1}_{\vartheta(b+c)>2a+b+c}, \tag{2.9}$$

where

$$w(a,b,c) = \min\left(\frac{5-3c}{8}, 1-2b\right)\mathbf{1}_{\min((5-3c)/8,1-2b)>(1+a)/2}.$$

This is Pascadi's (5.8), with its positive margin suppressed only for notation. Every credited finite box below has a strict guard; levels are reduced before applying the theorem. No term available only at a gating equality is credited.

The well-factorable support condition for a descending prime vector is

$$p_1 \cdots p_{j-1} p_j^2 \leq X^D \quad \text{for every } j. \tag{2.10}$$

Pascadi's Proposition 5.5 gives (2.6) for this support when the total exponent is below $\vartheta(t_1)$, or below $\Theta(t_1,t_2,t_3)$ if $t_1 \leq 1/4$ and at least three factors are present.

3 The prefixed coefficient adapter

Lemma 3.1 (Relative sieves with a retained prefix)**.** *Fix finitely many sufficiently thin exponent boxes for descending prime prefixes, bounded away from exponent zero. Construct the relative cofactor sieve by Iwaniec's factorable-remainder construction.*

For a two-prime prefix (a,b), a lower sieve at cofactor cutoff X^w and total exponent D is admissible if

$$D > \max(2a, a+2b), \qquad D > a+b+2w, \qquad \Theta(a,b,v) > D \quad (0 \le v \le w), \tag{3.1}$$

with uniform margins over the box. For a three-prime prefix (a,b,c), a relative upper sieve at total exponent D is admissible if

$$D > \max(2a, a+2b, a+b+2c), \qquad \Theta(a,b,c) > D. \tag{3.2}$$

The corresponding averaged kernels are (2.5). *Ordinary Bombieri–Vinogradov supplies the total-level $1/2-\varepsilon$ fallback without the anatomy conditions.*

Proof. Freeze the external prime boxes and use relative level X^D divided by the *largest* possible prefix product. This removes a dependence of the relative sieve level on individual prefix primes, at a main-term cost tending to zero with box width.

The Iwaniec remainder is not an arbitrary mask on a well-factorable sequence. It is a finite sum of separated boxes of large prime factors, convolved with a fundamental-lemma component supported below $X^{O(v)}$, for a sufficiently small fixed v. This follows from the construction in [2, Section 5, especially (5.5)–(5.17)]. Its main term is the usual lower or upper linear-sieve function, with a controllably vanishing discretization error.

Prepend the external two-prime vector to each lower cofactor box. Its first two well-support inequalities are exactly $2a < D$ and $a+2b < D$. Relative lower support supplies all subsequent inequalities in (2.10); the useful range of the lower main term has $D-a-b > 2w$. Every third large factor then has exponent $v \le w$, so the entire interval condition in (3.1) permits Proposition 5.5, followed by Proposition 4.4, on each separated box.

For an upper cofactor sieve with three external primes, the three initial inequalities in (2.10) are precisely those in (3.2). The same factorization argument then applies. The fundamental-lemma component is split among the available factors exactly as in the Iwaniec construction; its exponent cost is absorbed by choosing v much smaller than the fixed margin. Small or absent remaining factors are either covered by the same separated-box proof or have modulus below $X^{1/2-\varepsilon}$ and are covered by BV. For example, a valid two-prefix box has $a+b < (D+a)/2 \le 7/16$, so adjoining only the very small fundamental component remains below $1/2$.

Completion from convolutions to genuine primes. The preceding Type II factorization is supplemented by the full prime-identity reduction; it is not by itself a theorem about prime outputs. We use the reduction in [4, Sections 5 and 9], as in the proof of [3, Theorem 5.4]. First restrict the output variable to $P^-(m) \ge z_0 = X^{1/(\log\log X)^3}$; this changes the prime-weighted sum only through negligible prime powers. Apply Heath–Brown's identity with $k=3$ and dyadically localize its at most six factors. Choose a fixed $\xi > 0$ sufficiently small relative to every reserved factorization margin. If one factor has length in $[X^{2\xi}, X^{1/3+\xi}]$, Proposition 5.5 and Proposition 4.4 apply after grouping all other factors together. The rough Siegel–Walfisz sequences and the reduction of divisor-bounded coefficients to bounded coefficients are the same as in that prime-identity argument; their logarithmic losses are absorbed in the arbitrary logarithmic saving.

In the complementary ranges every Möbius factor has length below $X^{2\xi}$, since its length is at most $(2X)^{1/3}$. Each remaining smooth factor is either below $X^{2\xi}$ or above $X^{1/3+\xi}$, and at most two can be large. With one large smooth factor its length is at least $X^{1-10\xi+o(1)}$, and the pointwise fundamental-lemma estimate [4, Lemma 5.4] applies before summing the moduli. With two large smooth factors the product M of the other factors satisfies $M \le X^{8\xi+o(1)}$. The absolute-modulus divisor estimate [4, Proposition 5.2] applies because all our moduli satisfy $Q \le X^{5/8}$ and, for sufficiently small ξ,

$$Q \le (X/M)^{2/3-3\xi}.$$

There is no remaining three-large-factor case. The appended prime boxes and the small-prime component occur only in the modulus coefficient, independently of these Heath–Brown variables. For fixed Iwaniec parameters their coefficients are bounded by a parameter-dependent constant (or by a fixed logarithmic power after harmless subdivisions). Thus the pointwise estimate in the one-large-factor case and the absolute modulus sum in the two-large-factor case remain valid after inserting every such coefficient. This pays the complementary terms and establishes the actual prime progression estimate required for the prefixed sieve.

There are finitely many boxes for each fixed choice of parameters. Choose the arbitrary logarithmic saving in the progression theorem large enough before summing them. The pointwise cofactor sieve inequalities are applied first, and their errors are averaged over all prefix primes afterwards. This proves the averaged statement without a pointwise large-modulus PNT assumption. Smooth ψ and the Mangoldt prime weight follow by partial summation. Finally, $1/\varphi(pq\cdots)$ can be replaced by $1/(pq\cdots)$ with a power-saving error because all external factors exceed a fixed power of X. □

Scope note 3.2. The raw identities $\lambda_D^-(pc) = -\lambda_{D/p}^+(c)$ and $\lambda_D^-(pqc) = \lambda_{D/(pq)}^-(c)$ hold with their respective prefix-support conditions. They explain the signs but do not establish the factorable-remainder adapter. The preceding boxed construction is the necessary analytic step.

4 A switched upper bound that keeps the rough cofactor

Let ω be Buchstab's function: $\omega(u) = 1/u$ on $1 \le u \le 2$ and $(u\omega(u))' = \omega(u-1)$ for $u > 2$.

Theorem 4.1 (Source-preserving upper switching)**.** *Let $\mathcal{R}$ be a compact exponent region with*

$$\eta \le c \le b \le a < 1/3, \qquad \frac{1-a-b-c}{c} > 1,$$

where $\eta > 0$. Define

$$T_{\mathcal{R}}(X) = \sum_{\substack{p>q>r \text{ primes} \\ (\log_X p, \log_X q, \log_X r)\in\mathcal{R}}} \sum_{P^-(m)\ge r} \Lambda(pqrm+2)\psi(pqrm/X).$$

Then

$$\limsup_{X\to\infty} \frac{T_{\mathcal{R}}(X)}{\mathcal{N}_\psi(X)} \le \frac{10}{3} \iiint_{\mathcal{R}} \frac{\omega((1-a-b-c)/c)}{abc^2}\,\mathrm{d}a\,\mathrm{d}b\,\mathrm{d}c. \tag{4.1}$$

No primality condition on the cofactor m is assumed or removed from the original source; its actual roughness is retained.

Proof. Partition the prefix into thin boxes and write $M = qrm$, $N = p$. The actual coefficients are

$$\alpha_p = \mathbf{1}_{p \text{ prime in its box}}, \qquad \beta_M = \sum_{qrm=M} \mathbf{1}_{q,r \text{ in their prime boxes}} \mathbf{1}_{P^-(m)\ge r}.$$

Every prime factor of M exceeds X^η, so $\Omega(M) \le \eta^{-1} + O(1)$ and $\beta_M = O_\eta(1)$. The prime coefficient α satisfies the Siegel–Walfisz condition and the required rough support. In particular, the full coupled cofactor weight is legal in Proposition 4.4; it is not replaced by independent marginal densities.

Apply an upper sieve to the *output* $pM + 2$, with modulus level $D = X^{3/5-\varepsilon}$. Proposition 5.2 factorizes every upper-weight box for our Type II parameter $N = p$, since $X^\eta \le N < X^{1/3}$. Proposition 4.4 then estimates the actual residue $pM \equiv -2 \pmod d$. The boxed Iwaniec construction and its fundamental component are treated with the same strict margins as in Lemma 3.1. Product conditions, ordering boundaries and $\psi(pM/X)$ are handled by boxes and partial summation, with logarithmic losses absorbed in the arbitrary saving.

The comparison term initially contains $\mathbf{1}_{(pM,d)=1}/\varphi(d)$. Its difference from $1/\varphi(d)$ is negligible even after absolute values: for each input $n = pM$,

$$\sum_{(d,n)>1} \frac{|\lambda_d|}{\varphi(d)} \ll (\log X)^C \sum_{\ell \mid n} \frac{1}{\ell} \ll_\eta X^{-\eta} (\log X)^C.$$

The sum of input coefficients is $O_\eta(X)$, giving $O_\eta(X^{1-\eta}(\log X)^C)$. Thus $g(d) = 1/\varphi(d)$ for odd squarefree moduli is a valid comparison density.

Sieve the output by all odd primes below $\sqrt{3X}$. For $n \in [X, 2X]$, the surviving outputs are precisely primes, for sufficiently large X. By (2.4) and $sF(s) = 2e^\gamma$ in this range, the prime-output counting factor is

$$\left(\frac{2}{3/5 - \varepsilon} + o(1)\right) \frac{\mathfrak{S}_2}{\log X}.$$

The input comparison mass is independently evaluated by the PNT and the rough-number asymptotic:

$$\sum_{p,q,r \in \mathcal{R}} \sum_{P^-(m) \geq r} \psi(pqrm/X) = \frac{I_\psi X}{\log X} \left(\iiint_{\mathcal{R}} \frac{\omega((1-a-b-c)/c)}{abc^2} \, \mathrm{d}a \, \mathrm{d}b \, \mathrm{d}c + o(1) \right).$$

This is an upper comparison evaluation, not an assumption about prime-output distribution. Multiplication by $\log X + O(1)$ restores the true prime Mangoldt weights. Higher prime powers contribute $O_\eta(\sqrt{X} \log^C X)$ because the prefix multiplicity is bounded. First let $X \to \infty$, then let the sieve margins and box widths tend to zero. This gives (4.1).

Thin ordering-boundary strips are controlled by the same nonnegative switched estimate with ordering relaxed. Their comparison integrals tend to zero, so an unsieved error of size $X/\log X$ is never silently spent as $o(\mathcal{N}_\psi(X))$. □

5 The adaptive Buchstab theorem

Define

$$d_0(a) = \max\{4/7, \vartheta(a)\}.$$

For $0 < a \leq 1/4$ and $0 \leq c \leq b \leq a$, one has

$$\Theta(a, b, c) \geq d_0(a). \tag{5.1}$$

For $a \geq 1/7$, this follows from the $\vartheta(a)$ candidate. For $a \leq 1/7$, the candidate $w(a, b, c)$ is at least $4/7$ and its gate is satisfied (boundary equality is also supplied by $\vartheta(a)$).

For $0 < w \leq b$, put

$$d_*(a, b; w) = \inf_{0 \leq v \leq w} \Theta(a, b, v), \qquad M(a, b, c) = \max(2a, a + 2b, a + b + 2c). \tag{5.2}$$

The infimum is over the *entire interval*; replacing it by endpoint values is invalid. Set

$$d_3(a, b, c) = \begin{cases} \Theta(a, b, c), & \Theta(a, b, c) > M(a, b, c), \\ 1/2, & \text{otherwise}, \end{cases}$$

and define the safe upper kernel

$$K(a, b, c) = \min\left\{ \overline{F}\left(\frac{d_3 - a - b - c}{c}\right), \frac{10}{3}\omega\left(\frac{1 - a - b - c}{c}\right) \right\}. \tag{5.3}$$

The direct term is $+\infty$ if $d_3 \le a+b+c$. All uses below are on strict finite boxes, with the nominal levels reduced.

For an admissible w, namely

$$d_* > \max(2a, a+2b, a+b+2w),$$

define

$$C(a,b;w) = \frac{1}{w}\overline{f}\left(\frac{d_*-a-b}{w}\right) - \int_w^b K(a,b,c)\frac{\mathrm{d}c}{c^2}. \tag{5.4}$$

Let

$$L_0(a,b) = \frac{1}{b}\overline{f}\left(\frac{d_0(a)-a-b}{b}\right), \tag{5.5}$$

$$\kappa(a,b) = \max\{d_0(a), \vartheta(a+b)\}, \tag{5.6}$$

$$L(a,b) = \max\left\{L_0(a,b), 0, \frac{1}{b}\overline{f}\left(\frac{\kappa-a-b}{b}\right), \sup_{w \text{ admissible}} C(a,b;w)\right\}. \tag{5.7}$$

An inadmissible direct branch is discarded; if its lower function is positive, its prefix conditions follow from $\kappa > a+3b$ and $a \le 1/4$.

Theorem 5.1 (Finite adaptive recovery)**.** *Choose finitely many disjoint prime-exponent rectangles in*

$$1/16 \le a \le 1/4, \qquad 1/16 \le b < a,$$

and finitely many admissible stopping choices on each. Suppose a uniform lower bound $g_R \ge 0$ for $L - L_0$ has been proved on each rectangle R. Then

$$\liminf_{X\to\infty} \frac{A(X)}{\mathcal{N}_\psi(X)} \ge a_0 + \sum_R g_R \iint_R \frac{\mathrm{d}a\,\mathrm{d}b}{ab}. \tag{5.8}$$

No term outside these rectangles must be estimated anew: its complete baseline bound is retained.

Proof. Initially omit squarefreeness and use true prime outputs. Buchstab gives, for any fixed $y \le 1/7$,

$$S(1;X^\rho) = S(1;X^y) - \sum_{X^y \le p < X^\rho} S(p;p).$$

The first term receives the complete $4/7$ lower sieve. On $a \ge 1/7$, the one-prefix upper sieve uses $\vartheta(a)$; on $a \le 1/7$, it uses $4/7$. The latter localized use is justified by the same two-factor Iwaniec box construction, not by masking a raw well-factorable sequence.

Refine a negative one-prefix term by the exact identity

$$S(p;p) = S(p;X^v) - \sum_{X^v \le q < p} S(pq;q).$$

The baseline upper and child-lower main terms reconstruct the original upper main term, because

$$\frac{\mathrm{d}}{\mathrm{d}v}\left[\frac{1}{v}\overline{F}(K_0/v)\right] = -\frac{1}{v^2}\overline{f}(K_0/v-1), \qquad K_0 = d_0(a)-a. \tag{5.9}$$

Thus any proved improvement $L - L_0$ in the child lower bound saves exactly the integral in (5.8); there is no omitted negative complementary sector.

To establish the child lower bound, apply Buchstab again:

$$S(pq;q) = S(pq;X^w) - \sum_{X^w \le r < q} S(pqr;r).$$

Lemma 3.1 supplies the initial lower term uniformly over every possible third modulus factor $v \le w$. The same lemma or BV supplies the direct triple upper term, and Theorem 4.1 supplies its switched alternative. Using the smaller upper bound on each fixed exponent box proves (5.4). Nonnegativity, the baseline, and the direct two-prefix level give the maximum in (5.7). These are inequalities for averaged prefix boxes; the choices depend only on exponents.

It remains to identify the unchanged baseline. At $y = 1/7$, it is

$$\frac{7}{2}\log 3 - \int_{1/7}^{1/4} \frac{4}{a(1-a)}\,\mathrm{d}a - \int_{1/4}^{8/31} \frac{2}{a(1-5a/2)}\,\mathrm{d}a = a_0.$$

For $y < 1/7$, the additional starting term and subtraction cancel exactly by

$$\frac{\mathrm{d}}{\mathrm{d}y}\left[\frac{1}{y}\overline{f}(D/y)\right] = -\frac{1}{y^2}\overline{F}(D/y-1), \qquad D = 4/7. \tag{5.10}$$

Therefore $y = 1/16$ introduces no new baseline debit. Finally restore squarefreeness and Mangoldt outputs using Section 7. Their errors are $o(\mathcal{N}_\psi(X))$. □

6 The finite certificate

The accompanying self-contained Python program regenerates its interval tables and verifies (5.8). It requires Python and NumPy, no external data and no network. Its default calculation uses

$$\Delta a = \Delta b = 2^{-12}, \qquad \Delta c = 2^{-14}, \qquad \text{guard} = 2^{-30}.$$

The rectangles have $a_0 \ge 1/16$, $a_1 \le 1/4$, $b_0 \ge 1/16$, and $b_1 \le a_0$. Stops w are chosen from the c mesh in $[1/32, b_0]$. A lower bound for Θ on each full (a, b, c) box is computed by bounding every affine candidate and requiring its gate on the entire box. The minimum of all these bounds up to w bounds d_* correctly.

6.1 Rigorous sieve-function enclosures

For the function tables use mesh $h = 1/2000$, maximum argument 80, and 48-digit directed Decimal arithmetic. The cumulative quantities

$$U(s) = s\overline{f}(s), \quad V(s) = s\overline{F}(s), \quad W(s) = s\omega(s)$$

are nondecreasing, by their positive delay equations. For any nondecreasing H,

$$H(v_0)\log(v_1/v_0) \le \int_{v_0}^{v_1} \frac{H(v)}{v}\,\mathrm{d}v \le H(v_1)\log(v_1/v_0).$$

Starting from the exact initial functions in (2.1)–(2.2) and $W = 1$ on $[1, 2]$, this inequality propagates lower and upper bounds by the method of steps. Each logarithm is widened to adjacent Decimal values; each conversion to binary64 is widened again. Interval queries use cumulative endpoint bounds and prefix/suffix envelopes. They do not assume that the numerically calculated functions happen to be monotone.

6.2 Box lower bounds

For a rectangle $[a_0, a_1] \times [b_0, b_1]$, the initial lower coefficient uses

$$\frac{1}{w}\overline{f}_{\mathrm{lo}}\left(\frac{d_{*,\mathrm{lo}} - a_1 - b_1}{w}\right).$$

The upper tail is bounded by a positive Riemann sum: on a c interval $[c_0, c_1]$, use the lower direct-function argument, the upper Buchstab enclosure on

$$\left[\frac{1-a_1-b_1-c_1}{c_1}, \frac{1-a_0-b_0-c_0}{c_0}\right],$$

and the denominator c_0^{-2}. The tail is integrated to b_1, which overestimates each actual tail to b. The baseline is bounded above at a_0, b_0, since $d_0(a) - a$ is decreasing. Finally use

$$g_R = \max\{0, C_{R,\mathrm{lo}} - L_{0,R,\mathrm{hi}}, L_{\kappa,R,\mathrm{lo}} - L_{0,R,\mathrm{hi}}\}, \quad \iint_R \frac{\mathrm{d}a\,\mathrm{d}b}{ab} \geq \frac{\Delta a}{a_1}\frac{\Delta b}{b_1}.$$

All elementary arithmetic is rounded outward with binary64 `nextafter`; the exponent-grid affine calculations are exact dyadic operations. Positive cumulative sums are enlarged by a standard γ_n bound. The final lower sum is accumulated in directed 55-digit Decimal arithmetic.

Proposition 6.1 (Certificate output)**.** *The default finite certificate has* 91 577 *active rectangles and proves*

$$\sum_R g_R \frac{\Delta a}{a_1}\frac{\Delta b}{b_1} \geq 0.12201440615467618077\ldots > \frac{61}{500}. \tag{6.1}$$

The program asserts the strict rational threshold 61/500*. It is the default run, not a coarse diagnostic mesh, that certifies Theorem 1.1.*

The finite certificate proves only the numerical inequality in the already-established analytic theorem. The asymptotic prime-progression inputs and complete source adapter are furnished by the preceding proofs.

7 Cumulative error and comparison ledger

The following choices are made in order: fix the finite exponent rectangles and guards; choose the Iwaniec discretization and fundamental-lemma exponent much smaller than all guards; choose the arbitrary logarithmic saving; let $X \to \infty$; finally let the auxiliary discretization errors tend to zero. The certificate leaves a strictly positive reserve above 61/500.

Term	Bound and treatment
Progression remainders	$O_{\psi,\varepsilon,B}(X/(\log X)^B)$, for arbitrarily large fixed B, before the finite box sum.
Factorable-weight discretization	$o_{\varepsilon\to 0}(1)\mathcal{N}_\psi(X)$ after $X \to \infty$; the full small-prime component is retained.
Prefix density corrections	$O(X^{1-\eta}(\log X)^C)$ on each fixed compact prefix region.
Switched coprimality main correction	$O(X^{1-\eta}(\log X)^C)$, because all input factors exceed X^η.
Final squarefree restriction	$O_\psi(X^{1-\rho}\log X + \sqrt{X}\log X) = o(\mathcal{N}_\psi(X))$.
Higher prime-power outputs	$O_\psi(\sqrt{X}\log^C X) = o(\mathcal{N}_\psi(X))$, with bounded prefix multiplicity.
Smooth-weight and logarithm changes	Partial summation and $\log(n+2) = \log X + O(1)$; the latter costs $O_\psi(X/\log^2 X)$ in switched terms.
Unselected factor regions	Their original complete baseline bounds remain. No unproved cancellation is assigned to them.

For squarefreeness, if $n \asymp X$ is X^ρ-rough but not squarefree, some prime $\ell > X^\rho$ satisfies $\ell^2 \mid n$. Thus

$$\sum_{\substack{n\asymp X\\ P^-(n)>X^\rho\\ \mu^2(n)=0}} \Lambda(n+2) \ll \log X \sum_{X^\rho<\ell\leq\sqrt{2X}} (X/\ell^2+1) \ll X^{1-\rho}\log X + \sqrt{X}\log X.$$

The same argument handles repeated prefix factors in intermediate fixed-positive-exponent regions. For output prime powers, the number of integers $\ell^k \le 3X$, $k \ge 2$, is $O(\sqrt{X})$; bounded tuple multiplicities and logarithmic weights give the displayed error.

8 Deep pursuit, scope tests, and what is not counted twice

8.1 The direct two-prefix special case

Keeping only $\kappa = \max(d_0(a), \vartheta(a+b))$ already yields a positive genuine-source improvement. On $a \ge 1/7$, its gain region is

$$1/7 < a < 1/4, \qquad 0 < b < (1-4a)/3.$$

The corresponding integral is approximately 0.02330041. This is included in the stronger finite certificate; it is not added a second time.

8.2 Initial-upper reoptimization is already contained

Define $\Phi_a(w) = \inf_{b \le w,\, c \le b} \Theta(a,b,c)$. For every $b \le w$, one has $d_*(a,b;b) \ge \Phi_a(w)$. Whenever its lower function is positive, the relevant prefix conditions hold, and

$$L(a,b) \ge \frac{1}{b}\overline{f}\left(\frac{\Phi_a(w)-a-b}{b}\right).$$

Integrating the difference from L_0 and using (5.9) shows that the improvement already supplied by children $b < w$ is at least

$$\frac{\overline{F}((d_0(a)-a)/w) - \overline{F}((\Phi_a(w)-a)/w)}{w}.$$

This is exactly the apparent gain from upgrading the initial upper term. Thus that reoptimization does not supply an additional gain in the complete child functional. This is a bookkeeping result for this functional, not an impossibility theorem for other sieves.

8.3 Exploratory values are not the accepted decimal

Ordinary quadrature suggested approximately 1.02991 for the full unswitched functional and 1.03534 with switching. These are useful guides but are *not* the decimal claimed in Theorem 1.1. The accepted coefficient is the exact expression $a_0 + 61/500$. Improving the mesh alone cannot be represented as an arithmetic mechanism reaching 1.1.

8.4 Why complete lower level $5/8$ was not imported

The general lower-weight factorization obstacle survives the restriction that sieve primes lie below $X^{8/31}$. Even the weaker inequalities (4.16) proved inside Pascadi's Proposition 4.4 do not automatically cover every lower-weight modulus above level $4/7$.

For a precise example family, take three prime exponents $a > 1/7$ and an even number k of exponents $b = a/(k+1)$, at lower level $4a$, where

$$\left(14 - \frac{11}{25(k+1)}\right) a > 2.$$

All lower-support inequalities hold, all factors may lie below $X^{8/31}$, and the actual modulus exponent $4a - a/(k+1)$ exceeds $1/2$. Take the Type II exponent $\nu < a$ sufficiently close to a. No large factor can enter Q_1. If at most one of the three large factors enters Q_2, then $2\nu+u_2+2u_3 > 1$. If at least two enter Q_2, the weaker fourth expression is at least

$$\frac{39}{25}\nu + 12a + \frac{11}{25}kb > 2.$$

Thus this uniform factorization architecture fails. All failure inequalities can be made strict, and an arbitrarily small perturbation of the prime boxes makes the lower-support inequalities strict as well. The many factors are factors of the *sieve modulus*, before roughness is imposed; the final physical input's at-most-three-factor property does not remove them. The obstruction does not rule out a modified lower support, another distribution theorem, or cancellation between coefficient cells.

9 Further pursuit: modified lower support and its unpaid tail

This section records a separate, substantive attempt to reach 1.1. None of its exploratory decimals is included in the certified gain of Section 6.

Proposition 9.1 (Even-prefix pruning)**.** *Let $2\rho < \theta$ and start with the lower Rosser tree of level X^θ, with sieve primes below X^ρ. Choose a prefix-free collection of admitted even-length prime prefixes, and delete each selected prefix with all its descendants. The retained weights remain a pointwise lower-sieve minorant.*

For a fixed finite collection of exponent cells bounded away from zero, a first cut at an even prefix of length j, with $T_j = t_1 + \cdots + t_j$, has the following normalized density-main-term loss:

$$\int_{\mathcal{P}_j} \frac{\overline{f}((\theta - T_j)/t_j)}{t_1 \cdots t_{j-1} t_j^2} \, \mathrm{d}t_1 \cdots \mathrm{d}t_j. \tag{9.1}$$

This is a statement about the formal sieve main term. Its arithmetic use also requires a valid remainder estimate for the complete retained weights.

Proof. Use the ordered finite inclusion–exclusion tree. The condition $2\rho < \theta$ ensures that no one-prime prefix is omitted by the initial square condition. Every first omitted prefix in the original lower tree is even, as are all the additional first cuts. For multiplicative test values $0 \le h(p) \le 1$, the exact subtree omitted at an even prefix P contributes

$$h(P) \prod_{\substack{\ell < P^-(P) \\ \ell \mid n}} (1 - h(\ell)) \ge 0$$

to the full product. Summing over first omissions proves the minorant inequality. This argument does not assert that the new lower weight is pointwise smaller than the old one.

For the difference of the formal density main terms, the descendants of an admitted even prefix form the ordinary relative lower sieve at level $X^{\theta - T_j}$ and cutoff X^{t_j}. Its normalized kernel is $t_j^{-1}\overline{f}((\theta - T_j)/t_j)$. Multiplying by the prime-prefix measure gives (9.1). Prefix-freeness ensures that no descendant is subtracted twice. □

For example, deleting every admitted two-prime prefix with $t_2 > c$, where $c \ge (\theta - \rho)/3$ and $c \le \theta/4$, has exact density cost

$$I_2(c) = \frac{1}{\theta} \int_2^{\theta/c - 2} \overline{f}(s) \log(\theta/c - 1 - s) \, \mathrm{d}s. \tag{9.2}$$

Indeed the original integral is over $c \le b \le \theta/4$ and $b \le a \le \theta - 3b$; substitute $s = \theta/b - a/b - 1$, $v = \theta/b$, and change integration order. If $c \ge \theta/6$, the formula uses only the elementary part (2.2). This exact formula does not itself supply the loss for any other factor region.

9.1 A complete combinatorial guard

For a finite exponent vector $\boldsymbol{t}$, define $\mathcal{F}(\boldsymbol{t})$ to hold if its sum is at most $1/2$, or if every $\nu \in [0, 1/3]$ is covered by a three-bin assignment of its entries. If the bin sums are u_1, u_2, u_3, the strong system (2.6), before margin adjustment, gives the interval

$$\left[u_1, \min\left\{\frac{1 - u_2 - 2u_3}{2}, \frac{2 - 5u_2 - 2u_3}{2}\right\}\right].$$

Alternatively one may use the weaker sufficient system (4.16) explicitly proved inside Pascadi's Proposition 4.4; its upper endpoint is

$$\min\left\{\frac{1 - u_2 - 2u_3}{2}, \frac{50 - 93u_2 - 50u_3}{71}, \frac{50 - 11u_1 - 125u_2 - 50u_3}{39}\right\}.$$

Both predicates are preserved when entries are decreased or removed: retain the same assignment and every constraint becomes weaker.

At an admitted even prefix $(t_1, \dots, t_{2j})$, retain its subtree only if the *cloned* vector $(t_1, \dots, t_{2j}, t_{2j})$ passes this guard. Every retained even modulus is a subvector of its own passed guard. Every retained odd modulus of length at least three is obtained by decreasing the cloned final entry of the preceding even prefix's passed guard. The one-prime case is below $1/2$. Thus this is a complete combinatorial rule for all depths, not merely a test of selected counterexamples.

For an arithmetic implementation the inequalities must have uniform margins on an overlapping cover of the required Type II range, and the Iwaniec small-prime component must also fit. Equality at a limiting endpoint is not a distribution theorem. In particular, a numerical bound for this modified sieve requires the sum of *all first-cut losses*, not only depths two and four.

9.2 A missed deep family and a complete finite alternative

The vector

$$(.15, .15, .15, .013, .013, .013, .013)$$

is admitted by the lower support at $\theta = .624$ or $.625$, has total $.502 > 1/2$, and survives the tempting depth-four cut that requires $t_4 > (1/2 - t_1 - t_2 - t_3)/2 = .025$. Set $\nu = .149$. None of the three large entries can enter the first bin. If at least two enter the third bin, the second distribution expression is at least $1.048 > 1$; otherwise at least two enter the second bin, and even the weaker fourth expression is at least

$$\frac{39}{25}(.149) + 12(.15) + \frac{11}{25}(.052) = 2.05532 > 2.$$

Strict perturbations give an open exponent region. Thus the first shallow cut does not complete the source.

A fully paid finite alternative is to end a negative three-prefix branch with an upper sieve at an actually available total level D_{av}. Writing $T = a + b + c$, its entire density loss against the ideal level θ is

$$\frac{\overline{F}((D_{\mathrm{av}} - T)/c) - \overline{F}((\theta - T)/c)}{abc^2}.$$

One can use the available three-factor level, or BV when useful, and delete the preceding positive parent if that costs less. This includes every deeper sector. With the present three-factor

information, ordinary quadrature gives a coefficient near 1.0299, consistent with the adaptive theorem, rather than 1.1. The quadrature value is not claimed as a certified theorem.

The ideal complete lower main term at $\theta = 5/8$ would be

$$L(5/8) = \frac{16}{5}\log\frac{91}{64} = 1.126324554102970\ldots.$$

Therefore a modified-support proof of $a > 1.1$ needs a complete source-admissible construction whose *total* loss, including all depths, boundary boxes, margins and discretization, is strictly below

$$L(5/8) - 1.1 = 0.026324554102970\ldots.$$

Optimized low-depth experiments did not establish this total bound. There is no automatically summable tail: the elementary safe condition $e + 2f \le d$ after a passed prefix ending at d leaves an unsafe corner $e/d \to 1$, $f/d \to 0$, for which ignoring the factorization margin produces a logarithmically divergent crude majorant. A quantitative many-factor argument or a better complete finite termination remains necessary. This obligation is not hidden in the certified coefficient $1.020531534119\ldots$.

10 Matched budget, parity test, and conclusion

Using only the inherited comparison constant in (1.3), the matched budget is:

Lower coefficient a	Required upper coefficient e	Status
$0.898531534119\ldots$	$e < -0.276610077055\ldots$	Previous lower bound
$1.020531534119\ldots$	$e < -0.154610077055\ldots$	Proved here
1.1	$e < -0.075141611174\ldots$	Not proved here
$> 1.175141611174\ldots$	$e = 0$ would suffice	Not proved here

Every sufficient inequality requires a fixed positive margin and matched unbounded scales. The new source bound holds as a liminf on all sufficiently large scales, but no new statement about e is established.

At the matched population interface, the assignment

$$P_1 = 0, \qquad A = P_2 + P_3 = B_2 + B_3 = c_*\mathcal{N}_\psi(X), \qquad \mathcal{E}_{\mathrm{comp}} = 0$$

is consistent with the new lower inequality, because $c_* > a_*$. This is a consistency test of the proved scalar information, not a construction of an alternative sequence satisfying every prime-progression theorem. It shows why even a future complete equidistribution estimate would not by itself close the present budget. The estimate constrains the total physical rough-source population; it does not distinguish the sign of the prime-versus-composite discrepancy. Therefore this manuscript supports the current Route B target without superseding it and does not establish twin-prime infinitude.

What done. Proved the source-preserving adaptive lower-sieve theorem and switched upper kernel, and certified the rational improvement $61/500$, giving $a_* = 1.020531534119\ldots$ with all source corrections paid.

What remains. A further approximately 0.079468465881 is needed to reach 1.1, and 0.154610077055 to reach the zero-discrepancy threshold. The complete composite discrepancy and a positive terminal reserve remain open. G3+/G5 are not claimed; the requested stronger terminal acceptance is unmet.

Next frontier. Obtain a genuinely stronger complete lower-weight distribution or a fully controlled modified lower support with its entire main-term loss paid, or match this proved coefficient with new favorable Route B joint arithmetic. The finite certificate is complete; the remaining gap is arithmetic.

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